

DETECTION AND TRACKING PERFORMANCE OF DIFFERENT DETECTORS RESULTS ON GMOT40

Model	Size	Method	HOTA	AssA	DetA	MOTA	FPS
YOLOV5	S	FairMOT	63.88	58.30	70.01	81.52	53.13
		MHT*	61.84	55.30	69.15	81.05	8.45
		SORT	61.17	53.99	69.32	80.34	248.69
		ByteTrack	68.29	65.41	71.31	83.77	6.37
		SCGTrack*	68.32	65.31	71.49	83.70	6.34
		Hybrid-SORT*	69.17	66.97	71.45	83.96	60.58
		UCMCTrack*	62.72	56.94	69.08	81.78	16.30
		SUSHI	67.69	64.36	71.21	83.55	
		DiffMOT	68.08	65.00	71.29	83.64	
		MLHT (Ours)	71.11	70.73	72.03	84.79	7.91
	M	FairMOT	65.22	64.05	66.42	81.85	52.18
		MHT*	63.26	61.06	65.56	81.38	7.47
		SORT	62.66	59.74	65.73	80.66	244.49
		ByteTrack	69.40	71.14	67.72	84.10	5.77
		SCGTrack*	69.45	71.04	67.90	84.03	5.79
		Hybrid-SORT*	70.24	72.70	67.87	84.28	58.16
		UCMCTrack*	64.08	62.69	65.49	82.10	15.19
		SUSHI	68.84	70.09	67.63	83.88	
		DiffMOT	69.21	70.74	67.71	83.97	
		MLHT (Ours)	71.99	76.45	68.45	85.12	6.12
YOLOV8	S	FairMOT	65.78	58.55	73.93	84.44	53.20
		MHT*	63.54	55.08	73.31	84.04	8.44
		SORT	63.23	54.44	73.44	83.24	248.66
		ByteTrack	70.30	65.29	75.68	87.02	6.52
		SCGTrack*	70.24	65.12	75.75	86.86	6.44
		Hybrid-SORT*	71.03	66.70	75.64	87.13	60.73
		UCMCTrack*	64.60	56.78	73.49	85.04	16.49
		SUSHI	69.84	64.50	75.64	86.67	
		DiffMOT	70.10	65.04	75.54	86.78	
		MLHT (Ours)	73.33	70.76	76.28	87.82	7.98
	M	FairMOT	68.39	50.96	91.80	87.05	52.21
		MHT*	65.79	47.46	91.19	86.65	7.45
		SORT	65.39	46.82	91.33	85.86	244.40
		ByteTrack	73.51	57.78	93.52	89.60	5.90
		SCGTrack*	73.43	57.60	93.59	89.44	5.87
		Hybrid-SORT*	74.39	59.20	93.48	89.71	58.29
		UCMCTrack*	67.04	49.18	91.38	87.64	15.35
		SUSHI	72.97	56.98	93.48	89.25	
		DiffMOT	73.29	57.52	93.38	89.36	
		MLHT (Ours)	77.18	63.30	94.11	90.39	6.17
	S	FairMOT	65.89	58.39	74.34	82.09	51.81
		MHT*	63.61	55.29	73.20	81.63	8.96
		SORT	63.24	54.35	73.57	80.85	246.87
		ByteTrack	70.34	65.23	75.84	84.55	6.51
		SCGTrack*	70.30	65.24	75.77	84.43	6.17
		Hybrid-SORT*	71.26	66.83	75.98	84.76	59.77
		UCMCTrack*	64.59	56.79	73.45	82.56	16.39
		SUSHI	69.75	64.14	75.85	84.34	
		DiffMOT	70.10	64.81	75.84	84.33	

Model	Size	Method	HOTA	AssA	DetA	MOTA	FPS
YOLOX		MLHT (Ours)	73.31	70.67	76.37	85.42	7.20
	M	FairMOT	68.30	60.07	77.66	83.27	49.15
		MHT*	66.02	56.96	76.52	82.81	8.08
		SORT	65.64	56.02	76.89	82.03	243.09
		ByteTrack	72.79	66.92	79.15	85.72	6.18
		SCGTrack*	72.75	66.93	79.08	85.61	5.88
		Hybrid-SORT*	73.71	68.53	79.30	85.93	57.59
		UCMCTrack*	67.00	58.47	76.77	83.74	15.49
		SUSHI	72.19	65.83	79.16	85.51	
		DiffMOT	72.54	66.50	79.15	85.50	
		MLHT (Ours)	75.84	72.37	79.68	86.59	5.59
YOLO11	S	FairMOT	63.31	57.56	69.62	81.44	52.43
		MHT*	61.29	54.48	68.97	80.69	8.10
		SORT	60.64	53.32	68.96	80.04	237.78
		ByteTrack	67.24	63.39	71.33	83.64	7.05
		SCGTrack*	66.95	62.84	71.31	83.55	6.57
		Hybrid-SORT*	68.15	65.10	71.36	83.83	60.27
		UCMCTrack*	61.88	55.46	69.07	81.67	15.93
		SUSHI	66.48	62.11	71.17	83.28	
		DiffMOT	66.64	62.33	71.28	83.33	
		MLHT (Ours)	70.94	70.35	72.09	84.77	7.53
	M	FairMOT	63.95	57.95	70.57	82.47	52.64
		MHT*	61.72	54.70	69.64	81.94	7.16
		SORT	61.49	53.77	70.33	81.33	233.65
		ByteTrack	68.61	65.06	72.37	84.99	6.56
		SCGTrack*	68.20	64.52	72.10	84.69	6.18
		Hybrid-SORT*	69.25	66.29	72.33	85.06	59.80
		UCMCTrack*	62.70	56.10	70.07	83.02	15.53
		SUSHI	67.98	63.98	72.22	84.81	
		DiffMOT	68.11	64.19	72.27	84.91	
		MLHT (Ours)	71.25	70.16	72.87	85.83	5.83